

Radiation Communication

A black and white photograph of a wave crashing on a beach. The wave is in the foreground, with white foam and spray. The beach is visible in the background, curving along the shoreline. The sky is dark. In the bottom right corner, there is a blue dotted pattern.

How different technologies fall along the
electromagnetic spectrum

ELECTROMAGNETIC SPECTRUM



Technology uses frequencies throughout the electromagnetic spectrum to communicate



Visible Light

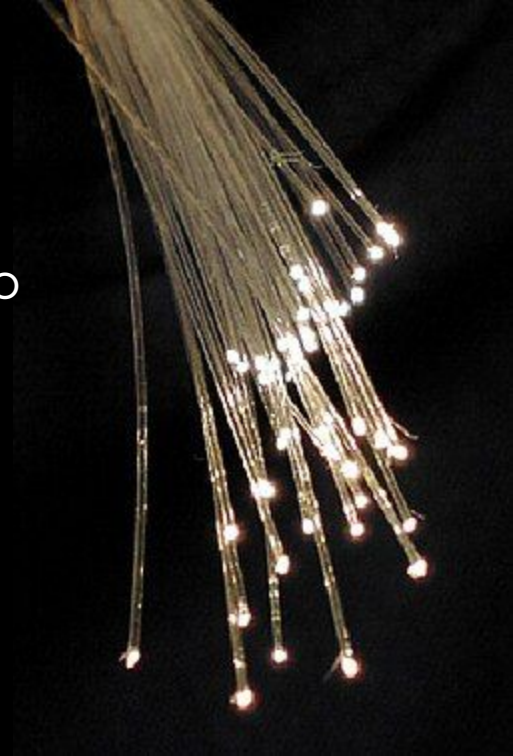


Visible light is the oldest means of communication using the electromagnetic spectrum



Infrared

Infrared is used by remote controls and fiber optics to communicate



Microwaves

Wifi, cellular and bluetooth communicate using microwaves.

Wifi: 2.4 GHz - 5 GHz

Bluetooth: 2.402 GHz - 2.480 GHz

Cellular: 600 MHz to 24 GHz

Microwave Oven: 2.45 GHz



Wi-Fi



Radio

NFC (Near Field Communication): 13.56 MHz



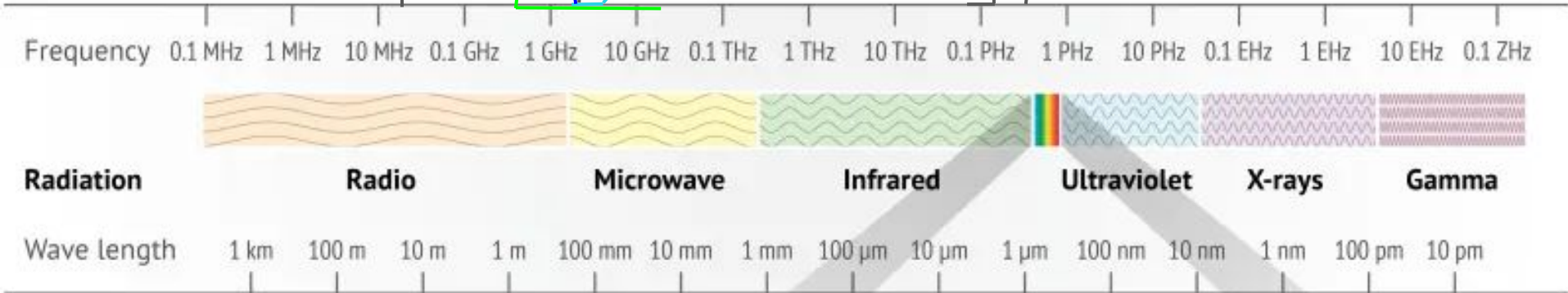
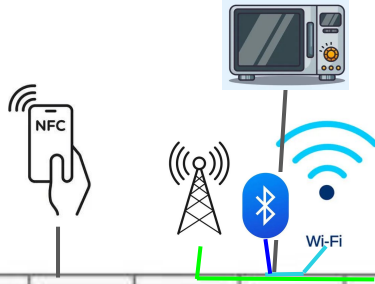
Technology on the Electromagnetic Spectrum

DANGER



**IONIZING
RADIATION
HAZARD**

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AC



Energy Levels

Wifi (router): 50 - 100 mW

Bluetooth: 0.01 mW - 100 mW

Cellular: 100 - 200 mW

NFC: does not apply

Microwave Oven: 600 W - 1200 W